

Curriculum Vitae

Amit Kumar Pal

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(Last updated: August 13, 2026)*

Designation: Associate Professor

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DOB: November 25, 1984

Nationality: Indian

Languages: English (fluent), Bengali (native), Hindi (intermediate)

Link to arXiv Preprints

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Link to Google Scholar Profile

I. PROFESSIONAL DETAILS

1. Associate Professor

Indian Institute of Technology Palakkad, India (June 2023 – Present)

2. Assistant Professor

Indian Institute of Technology Palakkad, India (September 2019 – June 2023)

3. Post-Doctoral Researcher

University of Warsaw, Poland (March 2019 – July 2019)

Mentor: Prof. Rafał Demkowicz-Dobrzański

4. Post-Doctoral Researcher

Swansea University, United Kingdom (October 2016 – March 2019)

Mentor: Prof. Markus Müller

5. Post-Doctoral Researcher

Harish-Chandra Research Institute, Prayagraj, India (March 2014 – October 2016)

Mentors: Prof. Aditi Sen(De) & Prof. Ujjwal Sen

II. ACADEMIC DETAILS

1. PhD in Physics

Department of Physics, Bose Institute, Kolkata, India (January 2009 – March 2014)

Supervisor: Prof. Indrani Bose

Thesis title: Ground and thermal state properties of model spin systems

PhD degree conferred by University of Calcutta, Kolkata, India

2. MSc in Physics

Department of Physics, University of Calcutta, Kolkata, India (August 2006 – August 2008)

3. BSc in Physics (Hons.)

Department of Physics, St. Xavier's College, Kolkata, India (August 2003 – August 2006)
BSc degree conferred by University of Calcutta, Kolkata, India

4. Higher-Secondary Examination

West Bengal Council of Higher Secondary Education (May 2001 – July 2003)
Hooghly Collegiate School, Chinsurah, India

5. Secondary Examination

West Bengal Board of Secondary Education (May 2001)
Hooghly Collegiate School, Chinsurah, India

III. RESEARCH INTERESTS

Quantum Information Theory; Interface of Quantum Information Theory and Quantum Many-Body Physics; Quantum Computing; Open Quantum Systems; Quantum Thermodynamics

IV. RESEARCH GRANTS

1. **Start-Up Research Grant (SRG)**, SERB, Govt. of India (December 2020 – December 2022)
2. **Core Research Grant (CRG)**, ANRF, Govt. of India (June 2024 – May 2027)

V. PUBLICATIONS

■ A total of 47 publications, including 41 published papers, 1 published book-chapter, and 5 pre-prints

■ Recently Published / Accepted Research Papers (2026)

1. **Developing a universal logical state-purification strategy**
C. B. Pushpan, T. K. Konar, A. Sen(De), and **A. K. Pal**
Phys. Rev. A 114, 022608 (2026)
DOI: [10.1103/k49w-qx4p](https://doi.org/10.1103/k49w-qx4p) [arXiv: [2502.01393](https://arxiv.org/abs/2502.01393)]
2. **Bridging the daemonic gap en route to charge multi-mode batteries via a single auxiliary**
C. B. Pushpan and **A. K. Pal**
Phys. Rev. A 114, 012607 (2026)
DOI: [10.1103/jr71-mkfl](https://doi.org/10.1103/jr71-mkfl) [arXiv: [2511.07243](https://arxiv.org/abs/2511.07243)]
3. **Simultaneous cooling of qubits via a quantum absorption refrigerator and beyond**
J. G. Krishnan, C. B. Pushpan, and **A. K. Pal**
Phys. Lett. A 591, 131871 (2026)
DOI: [10.1016/j.physleta.2026.131871](https://doi.org/10.1016/j.physleta.2026.131871) [arXiv: [2410.15871](https://arxiv.org/abs/2410.15871)]
4. **Scalable and deterministic Greenberger-Horne-Zeilinger state generation via graph states-assisted measurements**
Harikrishnan K. J. and **A. K. Pal**
Phys. Rev. A 113, 052450 (2026)
DOI: [10.1103/p2bp-mjk2](https://doi.org/10.1103/p2bp-mjk2) [arXiv: [2410.15892](https://arxiv.org/abs/2410.15892)]
5. **Scaling of entanglement entropy and correlations in the variable-range extended Ising model**
Harikrishnan K. J., D. Sadhukhan, and **A. K. Pal**
Phys. Rev. A 113, 052444 (2026)
DOI: [10.1103/vgrd-dg8j](https://doi.org/10.1103/vgrd-dg8j) [arXiv: [2504.01846](https://arxiv.org/abs/2504.01846)]

■ Complete list of publications can be found in [arXiv](https://arxiv.org/) or [Google Scholar](https://scholar.google.com/) profile.

VI. SUPERVISING EXPERIENCE

- **PhD:** 1 completed, 5 ongoing
- **MSc Thesis:** 9 completed, 1 ongoing

VII. SELECTED RECENT LECTURES / TALKS / CONFERENCE PRESENTATIONS

Note: Colloquium, lectures, invited seminars at conferences/symposiums/workshops, and invited technical talks in 2025-2026 only are listed.

■ Popular-level talk

1. Recent trends in quantum technology

- (a) March 2025: Popular-level lecture at the Department of Physics, Amrita University, Coimbatore, India

■ Colloquium

1. Harnessing localizable entanglement for quantum technology

- (a) November 2025: Colloquium at Department of Physics, University of Cyprus

■ Technical seminars

1. Benchmarking many-body quantum devices on the reciprocity landscape

Manuscript in preparation

- (a) July 2026: Invited seminar at the National Meeting on Quantum Information (MeetQIC) 2026 at Harish-Chandra Research Institute, Prayagraj, India

2. Universal logical state purification

Based on [Phys. Rev. A 114, 022608 \(2026\)](#) and [2512.09745](#)

- (a) February 2026: Invited seminar at the Department of Physics, SRM University, AP
- (b) December 2025: Invited talk at the international conference on “Quantum Foundations, Technologies and Applications” (QFTA2025) at IISER Mohali, Mohali, India
- (c) July 2025: Invited seminar at the National Meeting on Quantum Information (MeetQIC) 2025 at Harish-Chandra Research Institute, Prayagraj, India
- (d) April 2025: Invited seminar at the seminar series on “Advances in Quantum Theory and its Frontiers” at the Department of Physics, IIT (ISM) Dhanbad, India

3. How to cool a quantum bit?

Based on [Phys. Lett. A 591, 131871 \(2026\)](#), [Phys. Rev. A 107, 032602 \(2023\)](#), [Phys. Rev. A 105, 022214 \(2022\)](#), [Phys. Rev. A 104, 042208 \(2021\)](#), and [Eur. Phys. Lett. 125, 20007 \(2019\)](#)

- (a) June 2026: Invited seminar at the Department of Physics, Indian Institute of Engineering Science and Technology, Shibpur India

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